特集「インパクトアセスメント」

解 説

The EPS Default LCIA Method: A Method Based on External Costs

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Synopsis: The EPS system (EPS stands for Environmental Priority Strategies in product design) and its default LCIA method were introduced. The EPS system is mainly aimed to be a tool for a company's internal product development process, but may also have external external use like in environmental declarations. The EPS system is based on external environmental costs, and covers the following 5 impact categories; human health, production capacity of ecosystems, abiotic stock resource, bio-diversity and cultural and recreation values. This paper gives the scope and concept of the EPS system. An example how to calculate the damage cost for an emission is demonstrated and default values for damage costs for emissions and use of resources are given.

1. Introduction

The EPS system (EPS stands for Environmental Priority Strategies in product design) and its default LCIA method, which is based on external environmental costs, was developed to meet the requirements of an everyday product development process, where the environmental concern is just one among several others. The development of the EPS system was started during 1989 on a request from Volvo and as a co-operation between Volvo, the Swedish Environmental Research Institute (IVL) and the Swedish Federation of Industries. Since then it has been modified several times. The last modification was made within the Centre for Environmental Assessment of Products and Material Systems, CPM (Steen, 1999a and 1999b) in accordance with the ISO LCA standards, in particulate the ISO 14042 standard (ISO, 2000).

As a consequence of the requirement to adapt to an everyday product development environment, there is a demand on the use of a language that is easy to understand without too much training. Otherwise the risk of making a tool that may be good, but not used, is high. The following goal was formulated for the EPS system:

To be operative in a normal product-developing environment and to be able to assess which of two (or more) concepts that has the least impact on the environment. This means that the system must quickly be able to give recommendations in the early phases of the product development on the basis of general information. During later phases it shall allow more elaborate and precise recommendations and investigations as more detailed and specific information on the concepts become available. The demand about the system being operative contains a demand on usefulness and cost effectiveness. The extra efforts the designer makes are to result in a reasonable improvement for the environment and the product.

To assess the added value from all types of impacts. This requirement is partly a consequence of the demand on the system to be operative. It is considered unrealistic to take for granted that a product developer, who already has many technical and economical considerations to make, would be able to handle several different impact numbers. But he or she ought to have the possibility of choosing the degree of complexity and detail in the information.

To communicate an understanding of the magnitude of the impact. The result of the EPS analysis should be possi-

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ble to be weighed against other demands on the product.

To offer a forum for growth of a product related environmental strategy within a company in terms of "the 4 p's": plan, pattern, position and perspective. A plan is the original meaning of a strategy. A pattern means that it is not all decided from the beginning. A strategy grows as a pattern from various actions taken develops and many actors contribute. Position means that a company's environmental activities are profiled in relation to its market and competitors. Perspective means that it offers a way of learning.

When formulating a goal based on general information about a need and what might be possible to achieve, you make a choice. There are other options for the goal formulations. The choice of goal may be more restrictive, for instance the weighting against other non-environmental product properties is not normally subject to consideration in LCA. However, this is the reality for a product developer. In one way or another, it has to be decided how far to go in environmental improvement.

There is an ambition that the EPS system shall be able to give an answer within the order of 5 minutes in the first phases of the product development process and still meet all the requirements on transparency given by the ISO 14040-series. As the development process proceeds and the knowledge increases about whom that might deliver materials and which processes to use in manufacturing, the system must allow the exchange of more general data with more specific. A sensitivity analysis must be possible to make in order to guide the user of the system where to improve the product or the data used for the LCA.

The choice of 5 minutes as a target is based on experiences on what might be needed in a very early stage of the product development process. At that stage, several ideas may be tested and one or two may be chosen for further evaluation. 5 minutes is what it takes to describe the materials and processes of a life cycle of a concept that is not too complicated, like a hammer or a torch. Of course, 5 minutes will not do for a whole car or refrigerator, unless you have made a study before and know what the important parameters are. However, most complicated products are made up of components, and for these, the 5-minute target could be relevant. The time necessary for computation and data collection can be seen as negligible in this context, if adequate software and databases are available.

The EPS system is mainly aimed to be a tool for a company's internal product development process. It may be used externally and for other purposes, like for environmental declarations, for purchasing decisions and for education or for environmental accounting. But in those cases, the knowledge of the EPS system and its features and limitations is crucial.

The justification of many of the models used in impact assessments and for estimating inventory data relies on the fact that we analyse product systems. Such systems generally contain many emission or resource depletion events in various places, and we can get a fairly good estimate of the added impacts despite not knowing the individual impacts. Like for an aeroplane, the added weight of its next unknown 200 passengers may be estimated with higher relative precision than the weight of its next unknown passenger.

The models used may therefore not be applicable in other contexts. In particular, care should be taken when using the default models and data given in this work for specific impact assessment cases, like single plants or events.

2. Choice of default LCIA method

The EPS system in itself is flexible with respect to which LCIA method to use (Steen, 1999a). But it requires a default method to be available, so that the added impact from unit processes may be determined. This means that both a characterisation and a weighting method must be chosen.

Any choice is characterised by three things: what is included, how trade-offs are made and how uncertainty is handled. In the EPS default LCIA method, impacts are identified on a general level through five safeguard subjects: human health, bio-productivity, abiotic resources, biodiversity and cultural and recreational values. Impacts are considered equally as long as they

exist (=no discounting). The rational for that is partly the Rio convention that future generation shall have the same right to a healthy environment as we have and partly that the impact measure primarily is intended to communicate the size of the impact to the designer, not assess the tangible costs to the company or society. The geographical system borders for impacts are global.

Trade-offs may be made in several ways. The two extremes are based on rights and utilitarian approaches. In rights ethics, like in the ten amendments in the bible, or Islam's five duties, you need to fulfil a number of requirements, and the major concern is about doing so. In the LCA world this would correspond to the distance-to-target weighting. In the utilitarian approach, all impacts are measured in the same unit, like money. The EPS default impact assessment method use this type of approach. The rational for this is mainly that violation of human rights already are regulated by law. For instance, a local impact from an industrial plant on a single person may be insignificant in a monetary evaluation, but would be considered in a permit process. Another rational is that LCA as a method accepts a kind of utilitarian trade-offs in the LCI phase. Emissions are added together regardless of where they occur and regardless of their local impacts and violation of human rights. If the LCI phase accepts a utilitarian approach it is quite logical to use it in the LCIA phase too.

Uncertainty is a part of reality and is addressed in such a way that all quantities are given as a best estimate and as distributions. Distributions are described as lognormal and the standard deviation can thus be presented as a factor. The distributions are used in Monte Carlo simulations made to estimate uncertainties in the final LCA results, for instance to answer to what level of significance a design option A is better than B (Steen 1997).

3. Choice of impact categories and category indicators

When using a monetary approach, it becomes natural to select impact category indicators that have monetary value and which values are known to be significantly influenced by human activities. Besides they should also follow other requirements and recommendations by ISO (ISO, 2000), like comprehensiveness, no double counting and transparency. The category indicators used in the EPS default LCIA method (EPS 2000d) are shown below in Table 1 – 4.

3.1 Human health impact indicators

A wide spectrum of environmental impacts on human health is described in literature. Some of the impacts have known mechanisms and some is known from epidemiological studies.

Human health impact indicators may be chosen either to be numerous and very specific, or to be less and more general. The first may be tempting for the medical expert, but the information on corresponding cause-effects available today does not allow the use of diagnostic refinements more than in a few cases. There is also a limitation of studies of attitudes to many environmental related health effects. Besides if there are too many indicators, it will be difficult to obtain a good overview of the indicator results. This together speaks for a limited number of health impact categories. In Table 1 impact categories and category indicators chosen for human health are shown.

An alternative way of defining a health impact indicator is used by Goedkoop et al. (1997). They use WHO's concept of DALY, 'disability adjusted life years'. It covers

Table 1. EPS default impact categories and category indicators for human health effects

Impact category name	Category indicator name	Indicator unit	Notes
Life expectancy	Years of lost life, (shortname: YOLL)	personyear	Instead of excess mortality, which was used in earlier versions
Severe morbidity and suffering	Severe morbidity	personyear	Including starvation
Morbidity	Morbidity	personyear	Like a cold or flue
Severe nuisance	Severe nuisance	personyear	Would normally cause a reaction to avoid the nuisance
Nuisance	Nuisance	personyear	Irritating, but not causing any direct action

almost all health effects. For each type of health effect the degree of disability or similar is multiplied with its duration resulting in a figure corresponding to "lost person-years". There is an advantage in using the DALY system in that it gives an overview of all health effects. The reason for not adopting it at present as a default method for the EPS impact indicators is that it includes a large portion of subjectivity, which should belong to the weighting step. Another reason is that a practical degree of resolution when presenting a life cycle assessment in terms of category indicator results would contain about ten indicators. Health effects being perhaps the most important of effects should then be expressed in more indicators than the others. A closer look at what types of health effects that are related to environmental issues, we find cancer, respiratory effects, starvation, odour and soiling as common effects. It is regarded desirable to match these types of effects in the category indicators chosen.

3.2 Production capacity of ecosystems

Decreased yields of crop, fish&meat, wood and freshwater are end point effects associated with production capacity of ecosystems. Different types of crops are grouped together as they may be exchangeable as a source of carbohydrates. Different types of fish&meat may be exchangeable as a protein source. Different types of wood may be exchanged in most applications in a modern society. The category indicator chosen for these impact categories is a decreased production capacity of 1 kg. The weight refers to harvest weight for crop and fish&meat, while the dry substance weight is used for wood. Choosing dry weight basis for all three had given the most accurate measure, but normally dry weights are not available for crops or fish&meat while the forest industry often monitors the humidity of the wood it is buying.

The default impact categories and category indicators are summarised in Table 2

3.3 Abiotic stock resource indicators

3.3.1 Natural stock resources

Some authors have suggested impact indicators where resources are characterised due to their use to reserve ratio or due to their abundance (Lindfors et al., 1994). The reason for this is mainly to be able to highlight the use of scarce resources in an LCA, but it doubtful if this can be regarded as a characterisation based on natural science. It includes a large part of weighting when aggregating resources, many of which find very different use, in one measure. There is no common mechanism like in the formulation of the GWP that applies to the consequences of depletion of different resources.

In the choice of category indicators for the EPS default method, a resource, such as a metal ore is considered unique with regard to the metal. It is only exchangeable with regard to its concentration, chemical composition and location. The impact indicator is therefore defined as 1 kg of the resource in a reference state from which it is mined, i.e. in the state, which is normally referred to as reserves.

Table 2. EPS default impact categories and category indicators for ecosystem production capacity

Impact category name	Category indicator name	Indicator default unit	Notes
Crop production capacity	Crop production capacity (shortname: crop)	kg	Weight at harvest
Wood production capacity	Wood production capacity (short-name: wood)	kg	Dry weight basis
Fish&meat pro-duction capacity	Fish&meat production capacity (short-name: fish&meat)	kg	Full weight of animals
Base cat-ion capacity	Base cat-ion capacity	H^+ mole equivalents	Used only when models including the other indicators is not available
Production capacity for water	Production capacity for irrigation water (shortname: irrigation water)	kg	Must be acceptable for irrigation, e.g. with respect to persistent toxic substances
Production capacity for water	Production capacity for drinking water (shortname: drinking water)	kg	Fullfilling WHO criteria on drinking water (1997)

Land area is sometimes treated as a resource. However there is no change of global surface areas is possible due to human activities. It is only the quality of areas that can be changed. The quality is described by the impact categories: production capacity, bio-diversity and aesthetic values. The qualitative change between land/water may be described by those impact categories.

3.3.2 Anthropogenic stock resources

Buildings, machines, construction materials etc. represent values that can be destroyed by environmental impacts such as corrosion and soiling. There is no separate indicators formulated for these, but the impacts may still be accounted for in the impact modelling. For instance if there is a destruction of steel by corrosion from SO_2, impacts on the safe guard subjects are modelled through just including new pathways in the model. These pathways includes impacts from emissions and resource depletion when substituting and repairing the steel constructions

The default impact categories and category indicators are summarised in Table 3

3.4 Bio-diversity impact indicators

Part of the value of bio-diversity may overlap the safe guard subjects 'production capacity of ecosystems', and 'recreational and cultural values'. Here the safeguard subject 'bio-diversity' is focussing on the genetic resource values.

The most well known change in the safe guard subject bio-diversity caused by human activities is extinction of species. Each year a number of species is extinct. The value given to this change can be estimated from the cost of counteractive measures. Therefore it seems that the most reasonable option today is to use the yearly extinction of species as an indicator for the safe guard subject.

There is however a problem of finding the contribution to the yearly depletion from various activities. One reason is that it is not known exactly which species that are being extinct. The only way of estimating the contribution to species extinction is via the probability of extinction of red-listed species. Focussing on preserving red-listed species is claimed to be a good strategy for preserving other bio-diversity qualities as well.
So, the category indicator is defined as 'the normalised extinction of species'. The indicator unit is dimension-less. The category indicator name is shortened to NEX. The normalisation is made with respect to the species extinct during 1990, i.e. the sum of all indicator values on the globe 1990 is 1 NEX. If CO2 would double the extinction rate it would add 1 NEX extra per year.

The default impact categories and category indicators are shown in Table 4

3.5 Cultural and recreation value indicators

Changes in cultural and recreational values are difficult to describe by general indicators, as they are highly specific and qualitative in nature. Indicators are therefore defined when needed.

4. Characterisation

The characterisation step in a life cycle impact assess-

Table 3. EPS default impact categories and category indicators for abiotic stock resources

Impact category name	Category indicator name	Indicator default unit	Notes
Depletion of element reserves	= "element name" reserves	kg of element	E.g. Cu reserves, kg Cu
Depletion of fossil reserves	Natural gas reserves	kg	The hydrocarbon part
Depletion of fossil reserves	Oil reserves	kg	
Depletion of fossil reserves	Coal reserves	kg	
Depletion of mineral reserves	= "mineral name" reserves	kg	

Table 4. EPS default impact categories and category indicators for bio-diversity

Impact category name	Category indicator name	Indicator default unit	Notes
Extinction of species	Normalised extinction of species, short-name: NEX	Dimension-less	The normalisation is made with respect to the species extinct during 1990

ment aims at converting and possibly aggregating inventory results to category indicator results. For this it is necessary to find quantitative relations between the inventory parameters and impact category indicators.

4.1 Methods used for modelling

Characterisation factors express quantitative impacts on category indicators from elementary flows. The size of the impacts often depends on several pathways. Characterisation factors are therefore often a sum of several pathway-specific characterisation factors and each one of these is modelled separately.

There are in principle three types of models used to determine pathway-specific characterisation factors. The first type of methods we may call 'empirical', the second 'equivalency methods' and the third 'mechanistic'. The names are chosen because they represent carrying elements in the methods, not because they are purely mechanistic etc.

For the empirical method, the system boundaries in time and space is defined, and the characterisation factor for a substance is determined by dividing a category indicator value allocated to the system with an emission of the substance allocated to the system. The allocation of indicators and emissions may be necessary because of trans-boundary pollution.

The equivalency method use traditional equivalency factors like the global warming potential to calculate characterisation factors. For example to calculate CO's impacts on severe morbidity via global warming, the characterisation factor for CO_2 with respect to severe morbidity is used and just multiplied by the GWP for CO relative CO_2.

The mechanistic method typically estimates the portion of an emitted amount of a substance that will reach a sensitive target and use dose-response information to calculate the response per mass of the emitted substance.

4.2 An example: Assigning CO to impact categories and modelling the characterisation factor for CO with respect to severe morbidity

The EPS default LCIA method contains more than 2000 models. It is not possible to show them all here. They are described elsewhere by Steen (1999b). Here one example will be given: Assigning CO to impact categories and the determination of the characterisation factor for CO with respect to severe morbidity.

4.2.1 Definition of flow group:

Most of the CO emissions originate from gasoline driven cars and trucks. Some emission results from other sources with incomplete combustion like blast furnaces and uncontrolled wood fires.

The population of flows characterised is emissions to air, anyplace in the world 1990 and at any source strength.

4.2.2 Assignment to impact categories

Carbon monoxide is a well-known toxic gas. A large number of accidents and tests have revealed its toxicity (WHO, 1987). CO reacts with haemoglobin in blood and decreases its oxygen carrying capacity. At 10 -20 mg/m^3 there is a statistically significant decrease (3-7%) of the relation 'work time' to 'exhaustion' in healthy young men. At about 20 mg/m^3 as 8 hour mean (twice WHO recommended threshold limit value) there is statistically significant decrease of exercise capacity (=shorter duration of exercise before onset of pain) in patients with angina pectoris and increased duration of angina attacks. The first effect motivates an assignment of CO to moderate nuisance. It is hardly probable that the duration of severe exhaustion is influenced, because people tend to regulate their labour to a moderate degree of exhaustion. The second effect motivates an assignment to severe morbidity, a category to which heart attacks and pain must belong.

Carbon monoxide has also two other effects. It participates in photochemical reactions contributing to oxidant formation and it is a greenhouse gas, directly and indirectly through its reaction products. These mechanisms assign CO to all of the category indicators that CO_2 and ethylene is assigned to. (Table 5)

4.2.3 Characterisation of CO to air with respect to severe morbidity

<u>Definition of environmental system in which the impact is estimated</u>

CO as a molecule has a residence time in air of approxi-

Table 5. Assignment of CO to impact categories and selection of category indicators.

Pathway(s)	Impact category	Category indicator
Global warming	Life expectancy	YOLL
Direct exposure	Severe morbidity	Severe morbidity
Global warming	Severe morbidity	Severe morbidity
Global warming	Morbidity	Morbidity
Direct exposure	Nuisance	Nuisance
Global warming	Crop production capacity	Crop
Global warming	Wood production capacity	Wood
CO2 fertilisation	Wood production capacity	Wood
Global warming	Extinction of species	NEX

mately half a year. However rural background levels are in the order of 0.05 – 0.25 mg/m^3 which is several times below the WHO guidelines designed to protect from health effects (10 mg/m^3). This means that the direct impacts on health due to CO are mainly local, close to the sources. As the emissions and local exposure patterns considered is repeated on a global urban level the system chosen is urban and global and during the year 1990.

Model 1, direct exposure pathway

The characterisation factor is determined by an empirical method.

Category indicator value in system considered

1.2 billion people live in urban OECD areas and 1.4 billion in urban areas in non-OECD countries. WHO criteria for CO is estimated to be exceeded in almost 50% of the cities (GEMS, 1988). It is likely that only a part of the population in these cities is exposed. This part is assumed to consist of those who work or live permanently in the centre of the cities, roughly less than a third of the population, i.e. $4.3 \cdot 10^8$.

0.1% of the population which experiences concentrations above the WHO recommendations is assumed to be effected. This figure is chosen because the sensitive group is people suffering from cardiovascular diseases. As CV diseases account for almost a third of the deaths as a global average one can assume that the last percent of their lifetime is effected. Critical levels are estimated to be exceeded during 1% of the time (99%-ile or 10 cases per year of exceeding the 8-hour mean). This would mean $4.3 \cdot 10^8 * 0.001 * 0.01$ = 4300 person-years per year of severe morbidity.

Contribution to category indicators value from a flow unit

Global emissions of CO are 2600 million tons. Man made emissions account for 1600 million tons (UNEP, 1991) and dominate the urban exposure pattern. The contribution from 1 kg CO is at an average $1/(1600 \cdot 10^9) = 6.25 \cdot 10^{-13}$.

Calculation of pathway specific characterisation factor

The characterisation factor of CO for severe morbidity is thus $4300 * 6.25 \cdot 10^{-13} = 2.69 \cdot 10^{-9}$ person-years/kg for the direct exposure pathway.

Uncertainty

The uncertainty in the category indicator value is in the order of a factor of 10 as is the uncertainty in contribution from a typical product system CO emission. Mathematically the uncertainty is assumed to be represented by a log-normal distribution with a standard deviation corresponding to a factor of 5.

Model 2, global warming pathway

The characterisation factor is determined by an equivalency method using CO_2 as a reference.

Equivalency factor

GWP_{100} for CO is 3. (Houghton et. al, 1990)

Calculation of pathway specific characterisation factor

The characterisation factor of CO_2 for severe morbidity was determined in 3.1.4 to $3.53 \approx 10^{-7}$ person-years/kg CO_2. We thus obtain $3 * 3.53 \approx 10^{-7} = 1.06 \approx 10^{-6}$ person-years/kg CO for the global warming pathway.

Uncertainty

Indirect effects largely cause the GWP_{100} for CO. The GWP value 3 was presented in the first IPCC report but was withdrawn in later editions of IPCC assessments with reference to the large uncertainty involved. As CO will be oxidised to CO_2 in the atmosphere, it seems reasonable to assume that the GWP>1. On the other hand it is not as strong an oxidant precursor as most VOC's which were given the GWP 11. This indicates an uncertainty of a factor of three at most. In a log-normal distribution this would correspond to a factor of 1.7 as standard deviation. The uncertainty of the characterisation factor for CO_2 was determined in earlier (Steen, 1999b) to be represented by a log-normal distribution with a standard deviation corresponding to a factor of 3. The total uncertainty is therefore a log-normal normal distribution with a standard deviation corresponding to a factor of $\exp\sqrt{(\ln 1.7)^2 + (\ln 3)^2} = 3.4$

Calculation of characterisation factor

The characterisation factor for both pathways is thus $2.69 \approx 10^{-9} + 1.06 \approx 10^{-6} = 1.06 \approx 10^{-6}$ person-years/kg CO.

5. Weighting factors

A weighting factor is here defined as the ratio between the weighting indicator and the category indicator. The weighting indicator in the EPS 2000d method is ELU, (Environmental Load Unit) equal to one euro (€). The weighting factors will thus be the value of one category indicator unit in ELU or EURO. The reason for introducing ELU, was to emphasize the difference between normal tangible monetary flows and externalities. As for the modelling of characterisation factors, there is not room enough in this paper to describe all estimations of the weighting factor values, so only one example is given to illustrate how it is dine in practice. The estimation of all other weighting factors is reported in (Steen, 1999b).

5.1 An example: Weighting factor for life expectancy

To set a monetary value of human life is controversial, but nevertheless it is done indirectly in many of the decisions made by individuals and authorities. When building a road, the cost increases with increasing the safety. Somewhere the investment in safety has a limit. This may be described by the willingness to pay (WTP) for preserving lives. The same is valid for building hospitals. The more expensive and modern equipment and the more skilful personnel that is available the more lives may be saved.

The WTP for preserving lives in this way is normally referred to as VSL, the 'value of a statistical life'. It expresses the WTP for preserving lives given the prevailing circumstances in today's society. In the ExternE study (1995) various methods and studies for determining the value of a statistical life is reviewed (Table 6). ExternE conclude that the figure 2.6 million EUR (1990 years values) is a best estimate. In their conclusion they argue that the 'wage-risk' method is likely to give too low results while the 'CVM' method is likely to give too high results. The 'wage risk' method studies the relation of increased mortality and wages. The 'CVM' method is based on interviews and the 'market' method studies how prices on goods and services vary with safety.

In 1998 monetary values the ExternE 2.6 million EUR corresponds to $3.2 \cdot 10^6$ EUR. The variation in the study material is less than a factor of two, but there is en epistemological uncertainty indicated in the ExternE report (1995) which at least is of the same magnitude. Thus we assume a factor of three as being a reasonable estimate of the uncertainty range.

In the last decade it has become more and more common to express mortality impacts in terms of life expectancy. One reason is that in early studies, where the correlation between environmental pollution and mortality was discovered, daily mortality figures were correlated to daily environmental quality figures in a

Table 6. Results from various studies for the determination of the value of a statistical life (VSL). From ExternE (1995).

Method	European studies (Million EUR)	Studies in USA (Million EUR)
Wage-risk	2.8 - 3.5	3.5 - 5.5
CVM	4.1 - 6.3	1.4-2.5
Market	0.7 - 3.4	1.0 - 1.1
Average	2.5 - 4.4	2.0 - 3.0

particular area. In these studies it was not possible to say if a decreased environmental quality would shorten the life one day or much longer. In the last decade several studies has been made where different cities were compared and where remaining life expectancy was determined together with environmental quality measures. For particles in ambient air these impacts have been shown to be much more severe than those found in studies of short-term correlation.

The figures given above for VSL may be translated to life expectancy by looking at accidental mortality and assuming this to be randomly distributed in the population at risk. Even if it in reality is not randomly distributed, it may be argued that the people responsible for the safety decisions and thus the WTP figures hardly decide on the basis of skewed risk distributions.

The average life expectancy at birth in the OECD countries is around 75 years. Thus the average shortening of life due to random accidents are 37.5 years, which would give a value for the best estimate of WTP for one YOLL to 3200000/37.5 = 85000 EUR. This is the same value as used by ExternE, but recalculated to 1998 values.
The uncertainty range in the estimation of WTP for YOLL is estimated to a factor of five and it is assumed that it may be represented by a log-normal distribution with a standard deviation corresponding to a factor of two. (The uncertainty range ～ two standard deviations)

Choice of weighting factor

85000 ELU/YOLL with an uncertainty represented by a log normal distribution with a standard deviation corresponding to a factor of 2.

An overview of other weighting factors used in EPS 2000d are given in Table 1, appendix A

6. Estimation of external cost for an emission

Having determined the characterisation factors for emissions and resource flows for relevant category indicators, we may calculate the damage cost for each emission and resource flow unit. Such a calculation is demonstrated in Table 7. The emission is the same as in the characterisation example above.

An overview of damage costs for a number of emissions are shown in appendix A. There are a number of substances not present in the tables. The main reason is that only substances having a significant damage cost, as a global average, is listed. For others, the likely damage cost is zero.

7. Use of the EPS method in practice

The EPS method is mainly used in two ways. In conventional LCAs it is used as a weighting method. Characterisation is then made with midpoint indicators like GWP and acidification potentials. Then weighting is done directly from the inventory data using the EPS 2000 default indices/damage costs. The other way is to use databases with damage costs calculated for LCA modules like production of 1 kg of steel sheet, transporting 1 ton 1 km or incineration of 1 kg of PVC. In a practical design situation, there is an advantage of having

Table 7. Calculation of environmental damage costs for emission of a CO flow group unit

Sub-stance flow group	Indicator	Pathway	Pathway specific charcterisation factor	Uncer-tainty factor	Characterisation factor for all pathways	Indicators contribution to EPS default index	EPS default index (ELU/kg)
CO	YOLL	global warming	2.38E-06	3.4	2.38E-06	2.02E-01	
CO	severe morbidity	direct exposure	2.69E-09	5			
CO	severe morbidity	global warming	1.06E-06	3.4			
CO	severe morbidity	All			1.06E-06	1.06E-01	
CO	morbidity	global warming	1.96E-06	3.4	1.96E-06	1.96E-02	
CO	nuisance	direct exposure	2.50E-07	7	2.50E-07	2.50E-05	
CO	crop	global warming	2.27E-03	2.6	2.27E-03	3.41E-04	
CO	wood	global warming	-3.48E-03	2.4			
CO	wood	CO2 fertilisation	-3.93E-02	2			
CO	wood	All			-4.28E-02	-2.27E-03	
CO	NEX	global warming	3.78E-14	3.4	3.78E-14	4.16E-03	
CO	all	All					3.31E-01

external damage cost calculated for construction materials, manufacturing processes, transports, energy use and end-of-life processes. Then there is a direct feedback in the design situation and more creative solutions may be found compared to what will happen if you have to wait a week for the results.

8. Discussion

LCA as a method has developed along two lines: one based on natural science and one based on engineering science. The goal for the natural scientist has been to develop a theory as robust as possible. Their success criteria are about LCA being true and verifiable. The goal for an engineering scientist is that the product should be as environmentally good as possible. Their success criteria have more to do with LCA being useful. The EPS system is mainly an engineering tool. Therefore it has been natural to use less accurate characterisation models of the endpoint types in order to gain usefulness in the results.

A problem has been that users seldom understand characterisation modelling and weighting. Therefore there is a need for international bodies to criticize and create credibility for impact assessment methods.

Assessing environmental impacts are to a large extent dependent on ethical views. When presenting an LCIA method like in this paper it may seem as there is only one view. The EPS method was designed to allow companies to modify the default impact assessment method to better mirror the issues for them. This has happened a few times, but it is not common.

An important aspect is how to verify the models and calculations made. This has been done by comparing with other damage cost estimates and by comparing with the global GNP (Steen, 1999b). The evaluation shows a reasonable cost level, but of course, much is based on assumptions about the future and will not be possible to verify today.

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Appendix A

Table 1. Weighting factors and category indicators of the EPS default method v.2000.

Safe guard subject	Impact category	Category indicator	Indicator unit	Weighting factor (ELU/ indicator unit)	Uncer-tainty factor
Human health	Life expectancy	YOLL	Person-years	85000	3
Human health	Severe morbidity	Severe morbidity	Person-years	100000	3
Human health	Morbidity	Morbidity	Person-years	10000	3
Human health	Severe nuisance	Severe nuisance	Person-years	10000	3
Human health	Nuisance	Nuisance	Person-years	100	3
Ecosystem production capacity	Crop growth capacity	Crop	kg	0.15	2
Ecosystem production capacity	Wood growth capacity	Wood	kg	0.04	1.4
Ecosystem production capacity	Fish and meat production capacity	Fish and meat	kg	1	2
Ecosystem production capacity	Soil acidification	Base cat-ion capacity of soil	mole H+ -equivalents	0.01	2
Ecosystem production capacity	Production capacity for irrigation water	Irrigation water	kg	0.003	4
Ecosystem production capacity	Production capacity for drinking water	Drinking water	kg	0.03	6
Abiotic stock resources	Depletion of oil reserves	Fossil oil	kg	0.506	1.4
Abiotic stock resources	Depletion of coal reserves	Fossil coal	kg	0.049	2
Abiotic stock resources	Depletion of natural gas reserves	Natural gas	kg	1.1	2
Abiotic stock resources	Depletion of Ag reserves	Ag reserves	kg of element	54000	2.2
Abiotic stock resources	Depletion of Al reserves	Al reserves	kg of element	0.439	2
Abiotic stock resources	Depletion of Ar reserves	Ar reserves	kg of element	0	1
Abiotic stock resources	Depletion of As reserves	As reserves	kg of element	1490	2.2
Abiotic stock resources	Depletion of Au reserves	Au reserves	kg of element	1190000	3
Abiotic stock resources	Depletion of B reserves	B reserves	kg of element	0.05	10
Abiotic stock resources	Depletion of Ba reserves	Ba reserves	kg of element	4.45	3
Abiotic stock resources	Depletion of Bi reserves	Bi reserves	kg of element	24100	2.2
Abiotic stock resources	Depletion of Be reserves	Be reserves	kg of element	958	3
Abiotic stock resources	Depletion of Br reserves	Br reserves	kg of element	0	1
Abiotic stock resources	Depletion of Cd reserves	Cd reserves	kg of element	29100	2.2
Abiotic stock resources	Depletion of Ce reserves	Ce reserves	kg of element	45.2	3
Abiotic stock resources	Depletion of Cl reserves	Cl reserves	kg of element	0	1
Abiotic stock resources	Depletion of Co reserves	Co reserves	kg of element	256	3
Abiotic stock resources	Depletion of Cr reserves	Cr reserves	kg of element	84.9	3
Abiotic stock resources	Depletion of Cs reserves	Cs reserves	kg of element	512	3
Abiotic stock resources	Depletion of Cu reserves	Cu reserves	kg of element	208	3
Abiotic stock resources	Depletion of Dy reserves	Dy reserves	kg of element	1020	3
Abiotic stock resources	Depletion of Er reserves	Er reserves	kg of element	1410	3
Abiotic stock resources	Depletion of Eu reserves	Eu reserves	kg of element	3130	3
Abiotic stock resources	Depletion of F reserves	F reserves	kg of element	4.86	3
Abiotic stock resources	Depletion of Fe reserves	Fe reserves	kg of element	0.961	2.2
Abiotic stock resources	Depletion of Ga reserves	Ga reserves	kg of element	212	3
Abiotic stock resources	Depletion of Gd reserves	Gd reserves	kg of element	1060	3
Abiotic stock resources	Depletion of Ge reserves	Ge reserves	kg of element	2120	3
Abiotic stock resources	Depletion of H reserves	H reserves	kg of element	0	1
Abiotic stock resources	Depletion of He reserves	He reserves	kg of element	0	1
Abiotic stock resources	Depletion of Hf reserves	Hf reserves	kg of element	512	3
Abiotic stock resources	Depletion of Hg reserves	Hg reserves	kg of element	53000	2.2

Safe guard subject	Impact category	Category indicator	Indicator unit	Weighting factor (ELU/ indicator unit)	Uncer-tainty factor
Abiotic stock resources	Depletion of Ho reserves	Ho reserves	kg of element	4790	3
Abiotic stock resources	Depletion of I reserves	I reserves	kg of element	0	1
Abiotic stock resources	Depletion of In reserves	In reserves	kg of element	48700	3
Abiotic stock resources	Depletion of Ir reserves	Ir reserves	kg of element	59400000	3
Abiotic stock resources	Depletion of K reserves	K reserves	kg of element	0.01	10
Abiotic stock resources	Depletion of La reserves	La reserves	kg of element	92	3
Abiotic stock resources	Depletion of Li reserves	Li reserves	kg of element	0.1	10
Abiotic stock resources	Depletion of Lu reserves	Lu reserves	kg of element	11000	3
Abiotic stock resources	Depletion of Mg reserves	Mg reserves	kg of element	0	1
Abiotic stock resources	Depletion of Mn reserves	Mn reserves	kg of element	5.64	3
Abiotic stock resources	Depletion of Mo reserves	Mo reserves	kg of element	2120	3
Abiotic stock resources	Depletion of N reserves	N reserves	kg of element	0	1
Abiotic stock resources	Depletion of Na reserves	Na reserves	kg of element	0	1
Abiotic stock resources	Depletion of Nb reserves	Nb reserves	kg of element	114	3
Abiotic stock resources	Depletion of Nd reserves	Nd reserves	kg of element	115	3
Abiotic stock resources	Depletion of Ne reserves	Ne reserves	kg of element	0	1
Abiotic stock resources	Depletion of Ni reserves	Ni reserves	kg of element	160	2.2
Abiotic stock resources	Depletion of O reserves	O reserves	kg of element	0	1
Abiotic stock resources	Depletion of Os reserves	Os reserves	kg of element	59400000	3
Abiotic stock resources	Depletion of P reserves	P reserves	kg of element	4.47	3
Abiotic stock resources	Depletion of Pb reserves	Pb reserves	kg of element	175	2.2
Abiotic stock resources	Depletion of Pd reserves	Pd reserves	kg of element	7430000	3
Abiotic stock resources	Depletion of Pr reserves	Pr reserves	kg of element	471	3
Abiotic stock resources	Depletion of Pt reserves	Pt reserves	kg of element	7430000	3
Abiotic stock resources	Depletion of Rb reserves	Rb reserves	kg of element	27	3
Abiotic stock resources	Depletion of Re reserves	Re reserves	kg of element	7430000	3
Abiotic stock resources	Depletion of Rh reserves	Rh reserves	kg of element	49500000	3
Abiotic stock resources	Depletion of Ru reserves	Ru reserves	kg of element	29700000	3
Abiotic stock resources	Depletion of S reserves	S reserves	kg of element	0.1	5
Abiotic stock resources	Depletion of Sb reserves	Sb reserves	kg of element	9580	3
Abiotic stock resources	Depletion of Sc reserves	Sc reserves	kg of element	424	3
Abiotic stock resources	Depletion of Se reserves	Se reserves	kg of element	35800	3
Abiotic stock resources	Depletion of Sm reserves	Sm reserves	kg of element	632	3
Abiotic stock resources	Depletion of Sn reserves	Sn reserves	kg of element	1190	2.2
Abiotic stock resources	Depletion of Sr reserves	Sr reserves	kg of element	9.4	3
Abiotic stock resources	Depletion of Ta reserves	Ta reserves	kg of element	1980	3
Abiotic stock resources	Depletion of Tb reserves	Tb reserves	kg of element	5940	3
Abiotic stock resources	Depletion of Te reserves	Te reserves	kg of element	594000	3
Abiotic stock resources	Depletion of Th reserves	Th reserves	kg of element	288	3
Abiotic stock resources	Depletion of Ti reserves	Ti reserves	kg of element	0.953	3
Abiotic stock resources	Depletion of Tl reserves	Tl reserves	kg of element	3960	3
Abiotic stock resources	Depletion of Tm reserves	Tm reserves	kg of element	9900	3
Abiotic stock resources	Depletion of U reserves	U reserves	kg of element	1190	3
Abiotic stock resources	Depletion of V reserves	V reserves	kg of element	56	3
Abiotic stock resources	Depletion of W reserves	W reserves	kg of element	2120	
Abiotic stock resources	Depletion of Y reserves	Y reserves	kg of element	143	3
Abiotic stock resources	Depletion of Yb reserves	Yb reserves	kg of element	1980	3

Safe guard subject	Impact category	Category indicator	Indicator unit	Weighting factor (ELU/ indicator unit)	Uncer-tainty factor
Abiotic stock resources	Depletion of Zn reserves	Zn reserves	kg of element	57.1	2.2
Abiotic stock resources	Depletion of Zr reserves	Zr reserves	kg of element	12.5	3
Biodiversity	Species extinction	NEX	dimensionless	110.11E+	3

Table 2. External damage costs for inorganic substances

Substance flow group	Impact index, (ELU/kg)	Substance flow group	Impact index, (ELU/kg)
As	95.3	HCl	2.13
Benzene	3.65	HF	2.07
Butadiene	10.7	Hg	61.4
Cd	10.2	N2O	38.3
CH2O	6.47	NH3	1.96
CH4	2.72	Ni	0
CO	0.331	NOx	2.13
CO2	0.108	PAC (PAH)	64300
Cr	20	Pb	2910
Cu	0	PM10	36
Ethylene	3.45	Propylene	2.64
H2S	6.89	SO2	3.27

Table 3 External damage costs for organic halides

Substance flow group	Impact index (ELU/kg)	Substance flow group	Impact index (ELU/kg)
CFC:s		*Others*	
CFC-11	5.41E+02	HFC-23	1.34E+03
CFC-12	1.04E+03	HFC-32	6.42E+01
CFC-13	1.39E+03	HFC-43-10mee	1.77E+02
CFC-113	6.59E+02	HFC-125	3.54E+02
CFC-114	1.11E+03	HFC-134	1.33E+02
CFC-115	1.08E+03	HFC-134a	1.44E+02
		HFC-152a	1.55E+01
HCFC:s		HFC-143	3.21E+01
HCFC-22	1.94E+02	HFC-143a	4.87E+02
HCFC-123	1.23E+01	HFC-227ea	3.65E+02
HCFC-124	5.53E+01	HFC-236fa	8.85E+02
HCFC-141b	8.06E+01	HFC-245ca	6.75E+01
HCFC-142b	2.28E+02	SF6	2.76E+03
HCFC-225ca	2.13E+01	CF4	6.97E+02
HCFC-225cb	6.19E+01	C2F6	1.38E+03
		c-C4F8	1.01E+03
Bromocarbons		C6F14	7.52E+02
H-1301	2.20E+03		

Table 4. External damage costs for VOCs

Substance flow group	Impact index (ELU/kg)	Substance flow group	Impact index (ELU/kg)
Ethan	1.46E+00	1,2,3-Trimetylbenzene	2.41E+00
Propane	2.24E+00	1,2,4-Trimetylbenzene	2.38E+00
N-butane	2.15E+00	1,3,5-Trimetylbenzene	2.40E+00
I-butane	1.74E+00	O-ethyltoluene	2.23E+00
N-pentane	2.25E+00	M-ethyltoluene	2.28E+00
I-pentane	1.80E+00	P-ethyltoluene	2.28E+00
Hexane	2.57E+00	N-propylbenzene	2.07E+00
2-metylpentane	2.43E+00	I-propylbenzene	2.07E+00
3-metylpentane	2.32E+00	Methanol	1.44E+00
N-heptane	2.58E+00	Ethanol	1.95E+00
N-oktane	2.41E+00	I-propanol	1.46E+00
2-metylheptane	2.40E+00	Butanol	2.33E+00
N-nonane	2.29E+00	I-butanol	1.85E+00
2-metyloktane	2.36E+00	But-2-iol	1.66E+00
N-decane	2.45E+00	Acetone	1.46E+00
2-methylnonane	2.45E+00	Methyl ethyl ketone	1.85E+00
N-undecane	2.34E+00	Methyl i-butyl ketone	2.37E+00
N-dodecane	2.19E+00	Acetaldehyde	2.11E+00
Metyl-cyclohexane	1.87E+00	Propionaldehyde	2.33E+00
1-butene	2.59E+00	Butyraldehyde	2.30E+00
2-butene	2.57E+00	I-butyraldehyde	2.20E+00
1-pentene	2.46E+00	Valeraldehyde	2.26E+00
2-pentene	2.54E+00	Acrolein	3.32E+00
2-m-1-butene	2.40E+00	Methyl chloroform	1.15E+00
2-m-2-butene	2.84E+00	Allyl chloride	2.16E+00
Butylene	2.58E+00	Dimethylester	1.25E+00
Isoprene	2.11E+00	Dimethylether	1.66E+00
Acetylene	1.64E+00	Propylene glycol methyl ether	2.54E+00
Toluene	1.95E+00	Propylene glycol methyl ether acetate	1.70E+00
O-xylene	1.91E+00	ethylacetate	1.68E+00
M-xylene	2.20E+00	n-butylacetate	1.94E+00
P-xylene	2.25E+00	i-butylacetate	1.66E+00
Etylbenzene	2.11E+00	Average NMVOC	2.14E+00

Table 5. External damage costs for emissions for water

Substance flow group	Impact index, (ELU/kg)
BOD	2.01E-03
COD	1.01E-03
N-tot	-3.81E-01
P-tot	5.50E-02

Table 6. External damage costs for pesticides

Substance name	Impact index (ELU)/kg	Substance name	Impact index (ELU)/kg
Alachlor	3.57E-01	Fenamiphos	1.43E+01
Aldicarb	3.57E+00	Glyphosate	3.57E-02
Aldrin	1.19E+02	Heptachlor	7.13E+00
Atrazine	1.02E-01	Hexachlorbenzene	4.46E+00
Benomyl	7.13E-02	Lindane	1.19E+01
Captan	2.74E-02	Malathion	1.78E-01
Carbaryl	3.57E-02	Methomyl	1.43E-01
Carbofuran	7.13E-01	Methoxychlor	7.13E-01
Chlordane	7.13E+00	Naled	1.78E+00
Chlorpyrifos	1.19E+00	Oxamyl	1.43E-01
Cypermethrin	3.57E-01	Paraquat	7.93E-01
2,4-Dichlorophenoxyacetic acid (2,4-D)	3.57E-01	Permethrin	7.13E-02
Demeton	8.92E+01	Phosphine	1.19E+01
Dichlorvos (DDVP)	7.13E+00	Pirimifos-methyl	3.57E-01
Dieldrin	7.13E+01	Propachlor	2.74E-01
Diflubenzuron	1.78E-01	Resmethrin	1.19E-01
Dimethoate	8.92E+00	Sodium fluoracetate	1.78E+02
Diquat	1.62E+00	Thallium sulfate	4.46E+01
Disulfoton	8.92E+01	Thiram	7.13E-01
Endosulfan	5.94E-01	2,4,5, Trichlorophenoxyacetic acid (2,4,5-T)	3.57E-01
Endrin	1.19E+01	Warfarin	1.19E+01
		Zinc phosphide	1.19E+01

Table 7. External damage costs for land use activities

Activity	Unit	Index,(ELU/unit)
Arable land use	m2year	0.001562
Forestry	m2year	0.00055
Forestry	m3	6.25
Hardmaking of forest land	m2year	0.0455
Littering	m2	13.9